VisuLogic: A Benchmark for Evaluating Visual Reasoning in Multi-modal Large Language Models

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Abstract

Visual reasoning is a core component of human intelligence and a critical capability for advanced multimodal models. Yet current reasoning evaluations of multimodal large language models (MLLMs) often rely on text descriptions and allow language-based reasoning shortcuts, failing to measure genuine vision-centric reasoning. To address this, we introduce VisuLogic: a benchmark of 1,000 human-verified problems across six categories (e.g., quantitative shifts, spatial relations, attribute comparisons). These various types of questions can be evaluated to assess the visual reasoning capabilities of MLLMs from multiple perspectives. We evaluate leading MLLMs on this benchmark and analyze their results to identify common failure modes. Most models score below 30% accuracy—only slightly above the 25% random baseline and far below the 51.4% achieved by humans—revealing significant gaps in visual reasoning. Furthermore, we provide a supplementary training dataset and a reinforcement-learning baseline to support further progress. Code, data, and baselines are available at https://visulogic-benchmark.github.io/VisuLogic.

1 Introduction

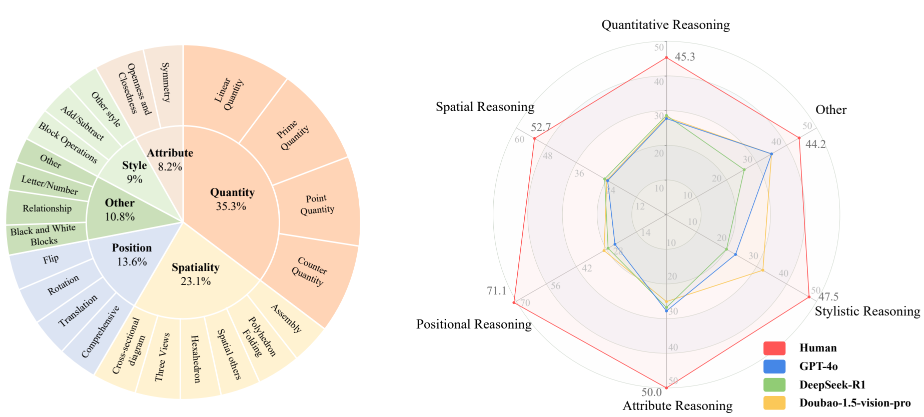

Figure 1: **Composition of the VisuLogic benchmark and performance of representative MLLMs.** The left figure shows the distribution of the 6 categories and their subcategories in VisuLogic. The right figure shows accuracies (%) achieved by MLLMs and by human on each category of VisuLogic.

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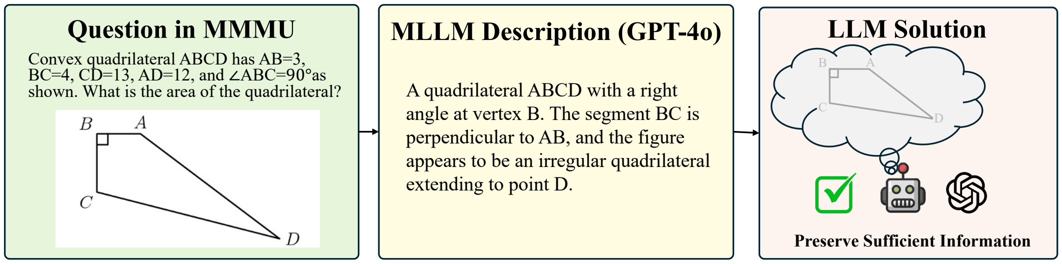
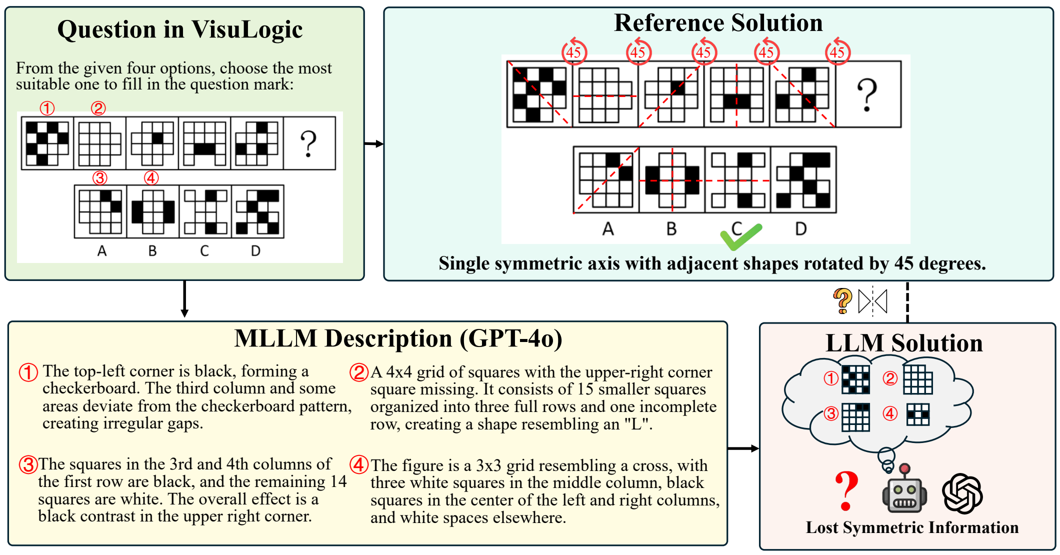


(a) Pipeline of "MLLM description→LLM" for Question in MMMU [89]. It is trivial that SOTA MLLMs extract key visual details, thereby enabling the LLM to answer questions solely based on language reasoning.



(b) Pipeline of "MLLM description→LLM" for Question in VisuLogic. Even SOTA MLLMs struggle to describe images precisely, leading to ambiguous interpretations.

Figure 2: **Comparison of the "MLLM description→LLM" pipeline on two benchmarks.** In MMMU, detailed descriptions lead to correct solutions, while in VisuLogic, critical visual cues (*e.g.*, symmetry, rotation) can be easily lost, causing the LLM to misinterpret the image. This highlights that textual reasoning alone is insufficient, underscoring the benchmark's demand for robust and in-depth visual reasoning.

Reasoning, as fundamental component of human intelligence, has become a critical criterion in evaluating progress toward Artificial General Intelligence (AGI) [26, 74]. Recent advancements in Large Language Models (LLMs) have demonstrated substantial improvements in reasoning capabilities across complex domains such as mathematics [61, 82, 81, 58], logical reasoning [68, 79, 23, 47] and coding [2, 35, 42, 32]. Techniques like Chain-of-Thought (CoT) [75] prompting and test-time compute scaling (e.g., OpenAI o1 [34] and Deepseek-R1 [18]) have significantly enhanced the reasoning performance of LLMs [18, 26, 74]. Along with the rapid development of language reasoning research for LLMs, considerable progress [84, 61, 58, 11, 50, 72, 51, 63, 73] has been made in improving multimodal reasoning capability of Multimodal Large Language Models (MLLMs).

These methods, which often incorporate reinforcement learning techniques [11, 50, 61] to enhance the reasoning capabilities of MLLMs, have achieved some early successes [84, 61, 58, 11, 50, 51, 63]. However, they typically rely on existing multi-modal benchmarks that struggle to accurately capture a model's core visual reasoning ability. For example, VLM-R1 [63] assesses "visual reasoning" with referring expression comprehension tasks [88, 55, 38], yet these tasks primarily focus on object localization, demanding only basic perceptual skills rather than more advanced visual cognitive processes. Meanwhile, several works [58, 61, 84] adopt mathematical problem-solving benchmarks that include diagrams—such as MathVista [52], MathVerse [91], and MathVision [69]—to evaluate visual reasoning. In practice, however, as [91] observes, many MLLMs translate these visual clues into textual descriptions and then rely on standard language reasoning. This approach can incorrectly

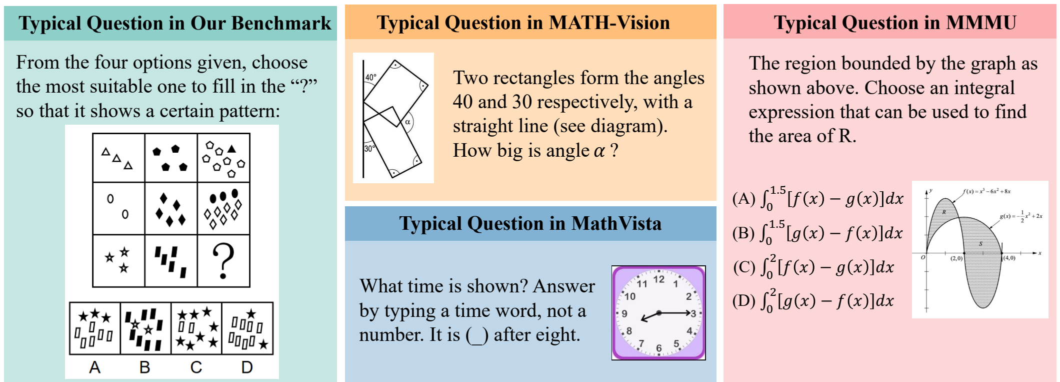


Figure 3: **Comparison of questions from different Benchmarks.** Compared to MathVista [52], MathVision [69], and MMMU [89], VisuLogic focuses more explicitly on assessing pure visual reasoning capabilities.

attribute language-driven results to visual reasoning, resulting in a misleading assessment of the model's visual reasoning capabilities [91, 30]. Consequently, designing new benchmarks that explicitly focus on vision-centric reasoning—rather than conflating it with text-based reasoning—remains critical for advancing MLLMs' visual reasoning capacities.

To address this limitation, we propose VisuLogic, a novel benchmark specifically designed to evaluate visual reasoning abilities in multimodal models without mixing them with purely text-based reasoning (see Figure 3). VisuLogic comprises carefully constructed tasks that span multiple reasoning categories (see Figure 1). As shown in Figure 5, these tasks are classified into six key types, such as Quantitative Reasoning, which requires understanding and deducing shifts in the quantity of certain elements within an image. In contrast to existing benchmarks, as demonstrated in Figure 2, state-of-the-art (SOTA) MLLMs often omit crucial visual details when describing VisuLogic problems, making it difficult for them to rely solely on a text-based inference shortcut. Indeed, even humans would find it challenging to capture every essential visual cue in a single description, so effectively tackling VisuLogic demands more robust, vision-centric reasoning. By reducing reliance on textual inference shortcuts, VisuLogic thus provides a stringent evaluation of MLLMs' genuine visual reasoning capabilities.

We conducted a comprehensive evaluation and systematic analysis to assess current models' visual reasoning capabilities. When leading text-only LLMs were supplied with detailed descriptions in place of raw images, their accuracy—Doubao-1.5-Pro (26.6%), Claude-3.7-Sonnet (25.9%) and Qwen2.5-72B-Instruct [83] (28.0%)—barely exceeded the random-chance baseline of 24.9%. This clearly demonstrates that textual reasoning alone are insufficient for solving our VisuLogic tasks. Even state-of-the-art multimodal arge language models (MLLMs)—including GPT-4o [33], Doubao-1.5-Vision-Pro, Gemini-2.0-Pro-Exp [64] and InternVL3-78B [94]—achieve only 26.3%, 28.1%, 28.0% and 27.7%, respectively, whereas human participants reached 51.4%. The substantial gap between these results and human performance underscores the challenge of robust visual reasoning in current MLLMs. Furthermore, we applied a simple reinforcement-learning (RL) fine-tuning step on our supplementary training dataset: this boosted the baseline model's accuracy from 25.5% to 31.1%, outperforming both open-source and closed-source counterparts. These findings illustrate the promise of the RL technique for strengthening MLLMs' visual reasoning capabilities.

In summary, our contributions are as follows:

- We propose a challenging visual reasoning benchmark that is inherently difficult to articulate using language, providing a more rigorous evaluation of the visual reasoning capabilities of MLLMs.
- We conduct comprehensive experiments to evaluate and analyze the benchmark, including extensive evaluations and comparative studies of various MLLMs under different setting.
- We identify the RL technique as a promising direction for improving the visual reasoning capabilities of MLLMs. Furthermore, we release both the training code and data to facilitate future research.